

Multi-Condition Diffusion for Paired Antibody Sequences

A decoding sweep reveals a quality–diversity Pareto frontier

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§1 Background and Motivation

Why BCR / TCR prediction?

- Paired heavy + light chain antibodies (VH+VL) target pathogens through dual-chain recognition
- Wet-lab discovery is slow and expensive (~\$2.6B and 10+ years per drug)

Why diffusion?

- ✓ Iterative refinement with global context at every step
- ✓ Native conditional control via classifier-free guidance



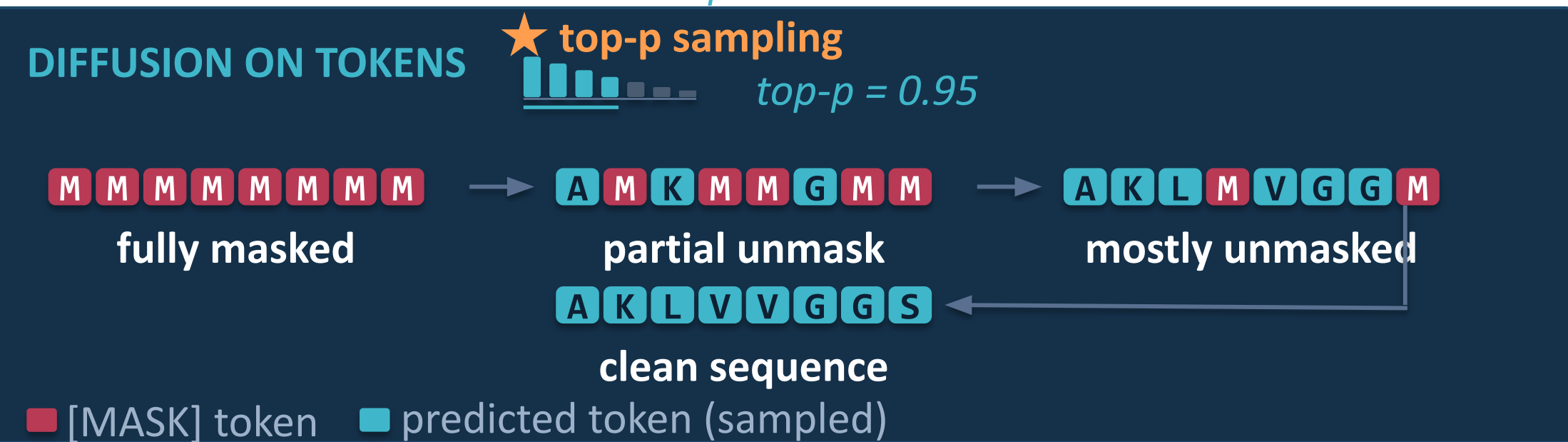
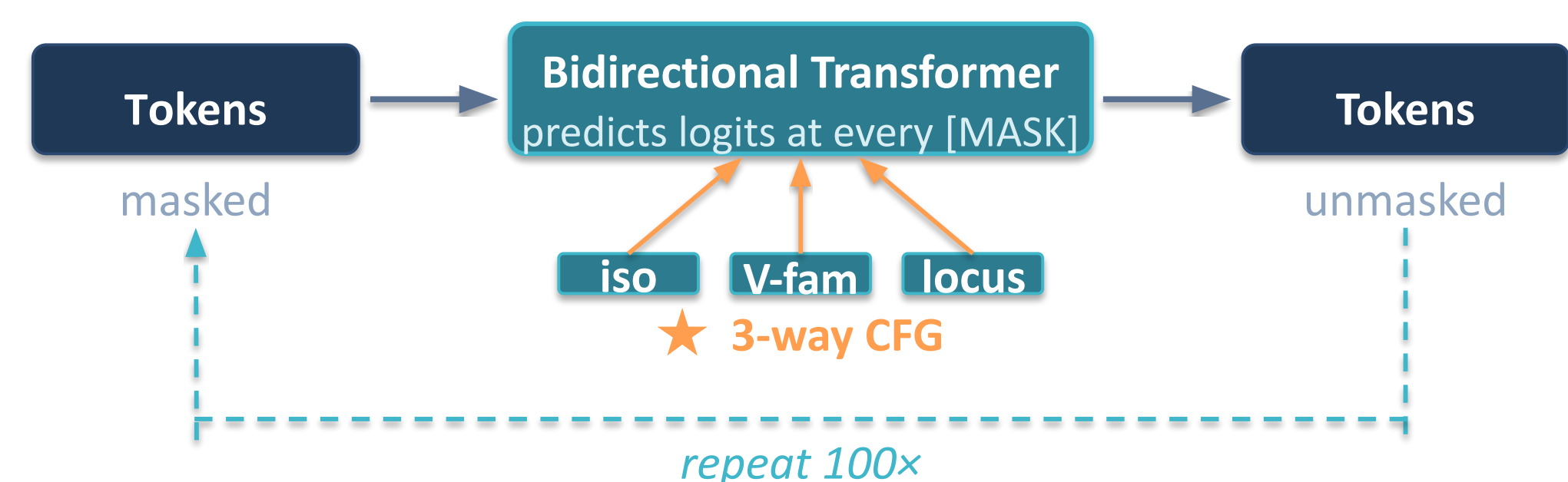
§2 Antibody Dataset

★ Dataset Not From Paper

- 2.17M paired human antibody sequences (BCR)
- MMseqs2 similarity reduction (95% / 90%) + 0% leakage

§3 Methodology - Two diffusion paradigms

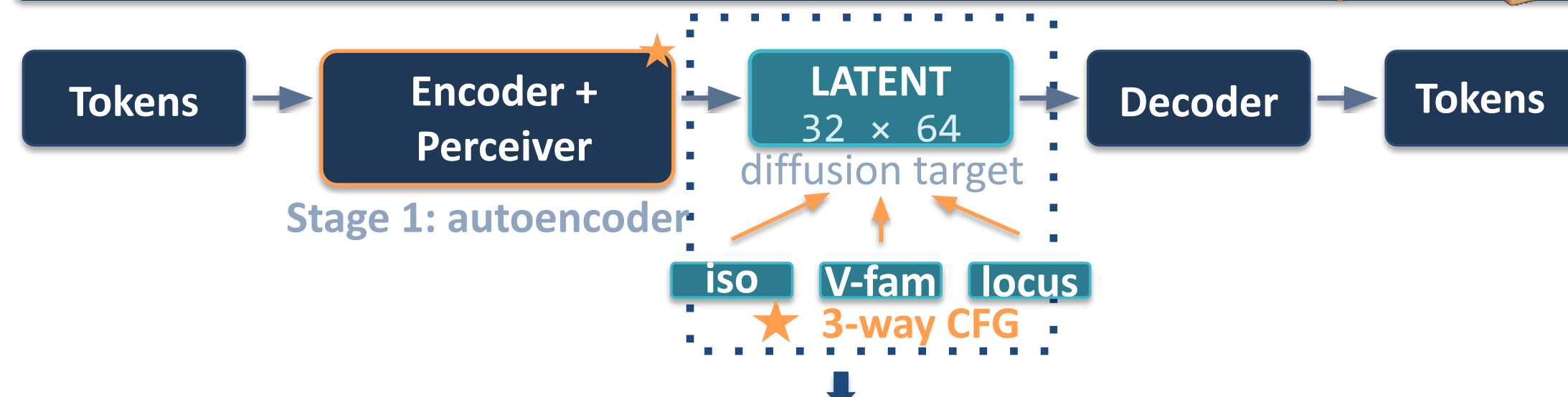
Paradigm 1 : Diffusion Protein Language Model (DPLM)



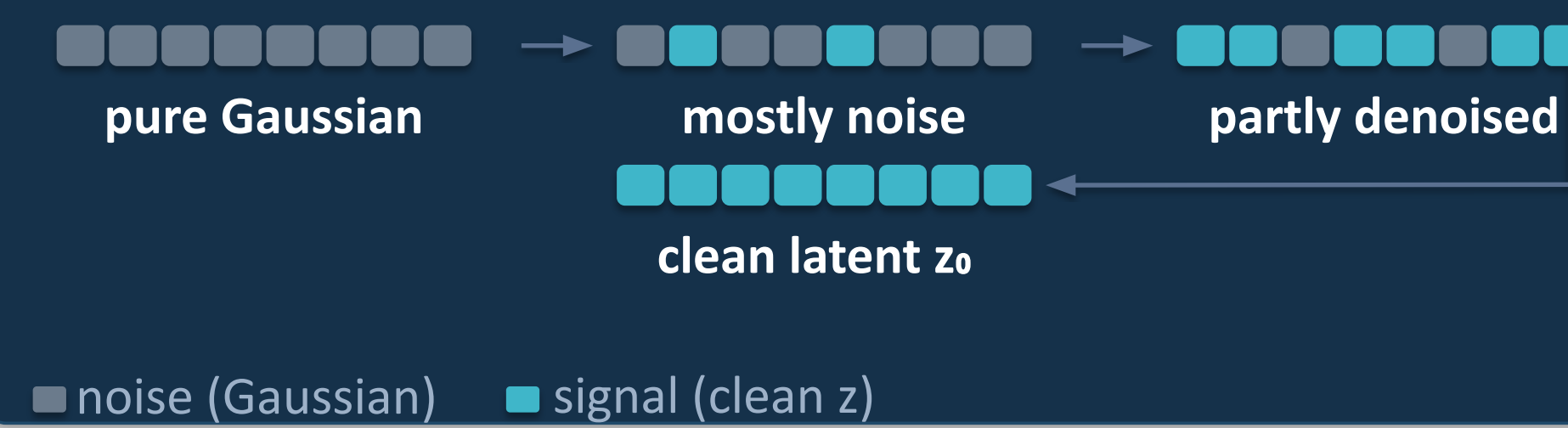
★ EXTENSIONS BEYOND PAPER

- 3-way CFG
- stochastic top-p decoder
- 20-config T × top-p sweep

Paradigm 2 : Latent Diffusion for Language Generation (LD4LG)



Stage 2: DIFFUSION ON LATENT



★ EXTENSIONS BEYOND PAPER

- AE trained from scratch
- 3-way CFG
- self-cond disabled (DDP)

§4.1 Results: Quality–Diversity Trade-off

Metric (<i>Bio/Actual Term</i>)	DPLM (default · T=1.0, p=0.95)	DPLM + sweep (tuned · T=1.3, p=0.99)	LD4LG
Format correctness (<i>linker recovery</i>)	100.0%	92.7%	99.7%
Sequence diversity (<i>4-gram diversity</i>)	0.051	0.207	0.136
Conditional accuracy (<i>V-family class</i>)	99.98%	93.8%	99.6%
Sequence recovery rate (<i>SRR</i>)	0.476	0.564	0.753
Model perplexity (<i>held-out NLL</i>)	1.37	1.37	1.08
Domain validity (<i>HMMER match rate</i>)	100%	99.5%	100%

*Best Score in Bold

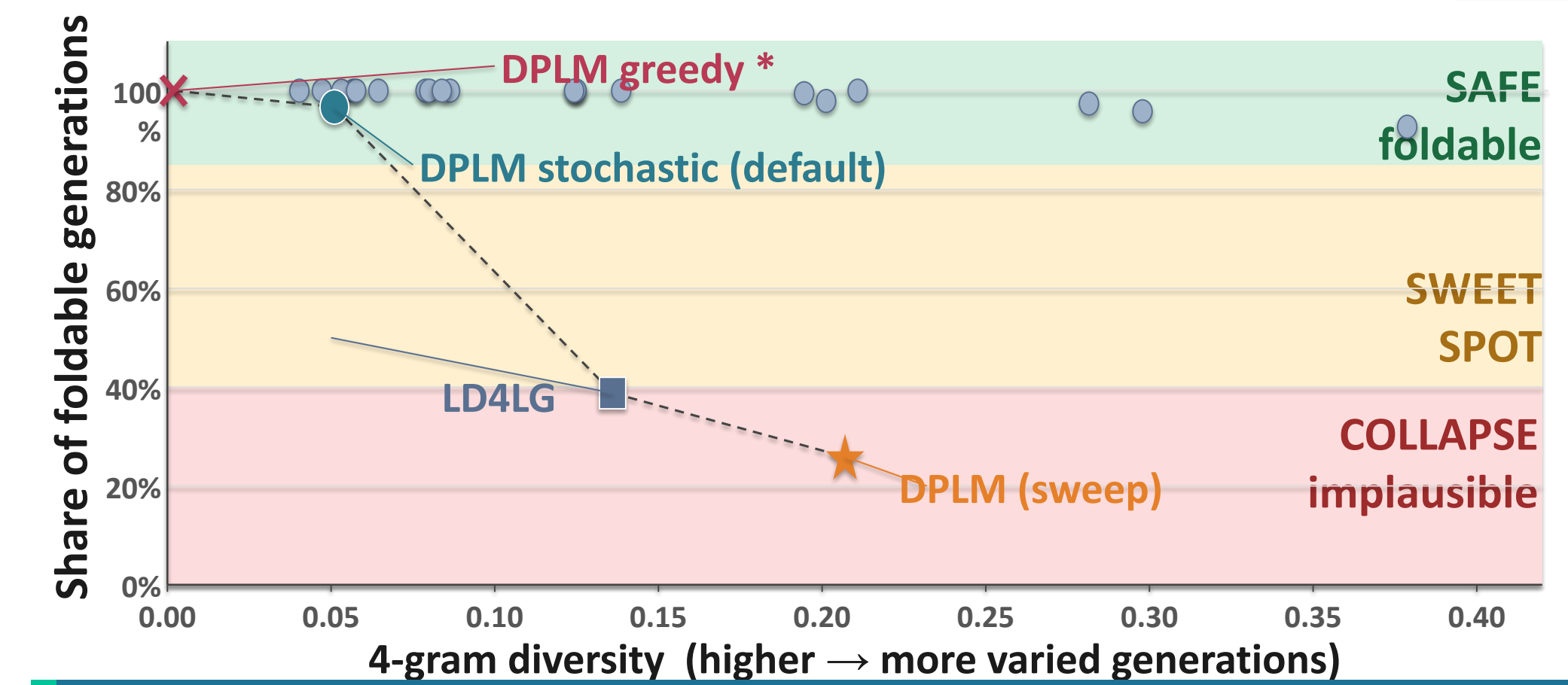
§4.2 Results: Foldable Share Across Models



97% → 26% : 71-pp drop in foldable share traded for 4× more diversity

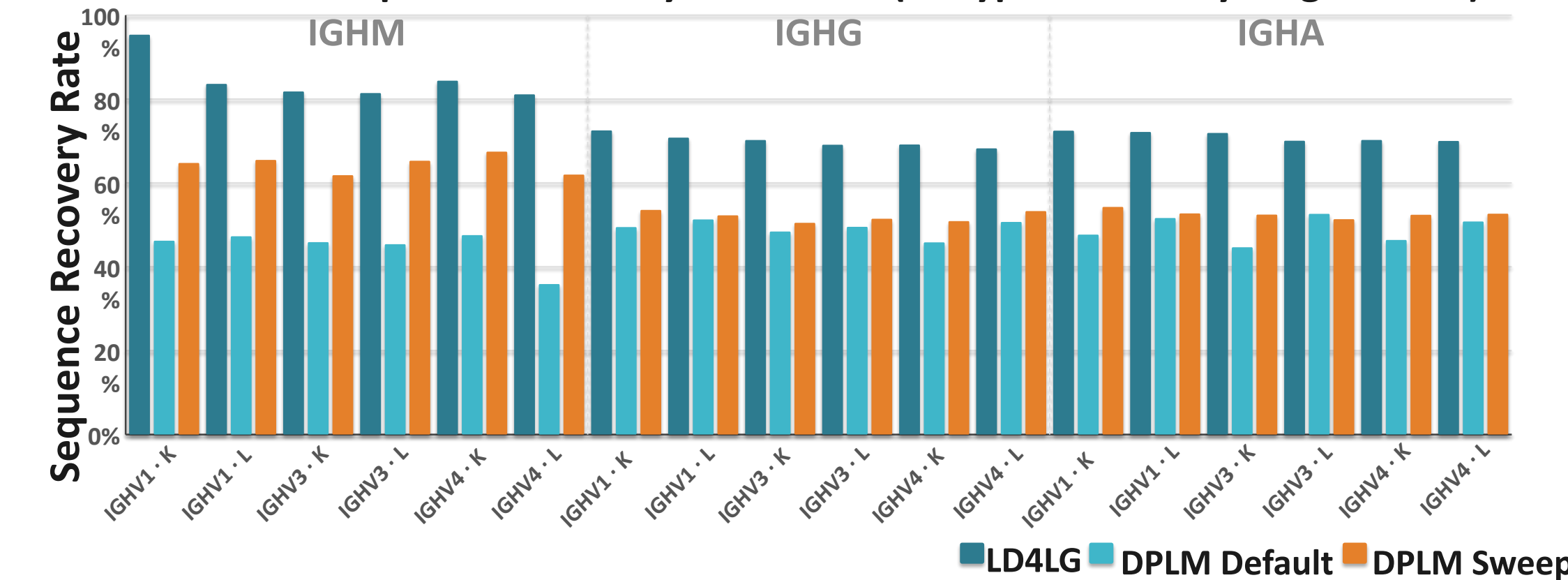
§4.3 Quality–Diversity Pareto

Diversity ↔ Foldability Trade-off



§4.4 Per-condition Recovery

Per-condition Sequence Recovery Rate — 18 (isotype × V-family × light-locus) cells



§5 Conclusion

- ✓ Decoding controls the quality–diversity trade-off
- ✓ DPLM is stable but lacks diversity
- ✓ CFG enables controllable generation

§6 Challenges + Future

Challenges:

- Balancing diversity and structural quality remains difficult
- Distributed training instability (DDP divergence) → limited single-GPU runs

Future directions:

- Better decoding strategies to improve the quality–diversity trade-off
- Hybrid approaches combining latent and token-space diffusion
- Extend to paired TCR (β + α) and cross species

★ EXTENSIONS BEYOND PAPER

§7 References

Lovelace et al. Latent Diffusion for Language Generation. NeurIPS 2023.
Wang et al. Diffusion Protein Language Models (DPLM). ICLR 2024.
Ho & Salimans. Classifier-Free Diffusion Guidance. NeurIPS-W 2022.
Salimans & Ho. Progressive Distillation. ICLR 2022.

Ruffolo et al. IgFold. Nature Communications 2023.
Olsen et al. OAS: Observed Antibody Space. 2022. opig.stats.ox.ac.uk/webapps/oas.
Bao et al. All Are Worth Words: A ViT Backbone for Diffusion (U-ViT). CVPR 2023.
Steinegger & Söding. MMseqs2. Nat Biotechnol 35, 1026 (2017).

Eddy. Accelerated Profile HMM Searches (HMMER3). PLOS Comp Biol 7, e1002195 (2011).